Example: Conversion of the decimal number 548 to its equivalent excess-3 code.

Decimal number	5	4	8
Add 3 to each bit	+3	+3	+3
Sum	= 8	7	11

Hence, the equivalent excess-3code 1000 0111 1011

The representation of Excess-3 code for the decimal numbers is given in the Table 8.5.

Table 8.5: Excess-3 Code of Decimal Number

Decimal Number	Excess-3 Code
0	0011
1	0100
2	0101
3	0110
4	0111
5	1000
6	1001
7	1010
8	1011
9	1100

8.6.4 Binary to Gray Conversion

To convert a given binary number to its equivalent gray code, the following rules are applied.

1. The MSB of the gray code is same as the MSB of the binary.
2. Coding starts from left to right, add each adjacent pair of bits to get the next bit of the gray code. Omit the carry, if occurs.

Example: Conversion of the binary number 1011 to gray code.

Step1: The MSB in gray code is same as the MSB of the binary

1 0 1 1 Binary
↓
1 1 Gray

Step2: Add the left most bit to the adjacent one.

1 + 0 1 1 Binary
↓
1 1 Gray

Step3: Add the next adjacent pair.

1 0+ 1 1 Binary
↓
1 1 1 1 Gray

Step4: Add the next adjacent pair and omit the carry.

1 0 1+ 1 Binary
↓
1 1 1 0 Gray

Hence, $(1011)_2 = (1110)_G$

Suffix 'G' is used to represent the Gray code.

8.6.5 Gray to Binary Conversion

To convert a given Gray code number into equivalent binary, the following rules are applied.

1. The MSB of the binary is same as the MSB of the Gray.
2. Coding from left to right, add the binary digit generated to the adjacent gray bit to get the next bit of the binary. Omit the carry if occurs.

Example: convert the gray code 1110 to its equivalent binary.

Step1: The MSB in binary is same as the MSB of the gray

1 1 1 0 Gray
↓
1 Binary

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